A formal proof of the finite-frame splitting bound

Conjecture 00000000951

quantumtool

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Statement and scope

The repository states: "The number of ways to split a frame into two Riesz sequences is at most 2^{n-1} (an upper bound on finite-dimensional splittings)."

We formalize the standard finite-family interpretation: the frame has n indexed vectors, and a split is an unordered pair of complementary subfamilies. For $n \geq 1$, this is the usual meaning of a partition of the frame indices into two parts. The exponent is $n-1$ because the two parts are unordered. If the intended parameter n is instead the dimension of the ambient space, or if swapping the two parts gives a distinct split, the statement needs that convention stated; this proof does not claim those variants.

Counting all splits

Let the frame indices be $\{0, 1, \ldots, n-1\}$. In any unordered split, exactly one of the two parts contains index 0. Call that part the first part. Each of the other $n-1$ indices can independently be placed in the first part or the second part. Hence the canonical representatives of all unordered splits are in bijection with the Boolean words of length $n-1$. There are exactly 2^{n-1} such words.

A split into two Riesz sequences is one of these unordered splits with an additional property. Any such condition can remove possibilities, but cannot increase their number.

Theorem 1. *For a frame with $n \geq 1$ indexed vectors, the number of unordered splits for which both subfamilies are Riesz sequences is at most 2^{n-1}.*

Proof. The canonical encoding above injects every qualifying split into the set of Boolean words of length $n-1$. The latter set has cardinality 2^{n-1}, proving the bound. □

Lean correspondence

The Lean project represents a canonical split by a Boolean word of length $\mathrm{Nat.pred}(n)$: index 0 is fixed to one side and each remaining bit records the side of one frame index. `allBitWords` recursively lists every Boolean word at a given length. The project proves that this enumeration is sound, complete, duplicate-free, and has length 2^m.

`qualifying_splits_bound` proves that filtering this enumeration by any Boolean predicate leaves at most 2^m entries. `riesz_split_bound` applies that theorem with the predicate for qualifying Riesz splits. No Riesz-sequence theorem is assumed or needed for this counting bound.

The package uses Lean 4.33.1 and `Std` only. `Check.lean` prints the dependency audit for every theorem, and no `sorry`, `native_decide`, or custom axiom occurs in the proof source.